

Topic 1

Radiation in treatment and medicine

- 1.1 Demonstrate an understanding of the methods that medical physicists can employ to help doctors solve medical problems, including:
 - a CAT scans
 - b ultrasounds
 - c endoscopes
 - d ionising and non-ionising radiation
- 1.2 Use the word 'radiation' to describe any form of energy originating from a source, including both waves and particles
- 1.3 Demonstrate an understanding that the intensity of radiation will decrease with distance from a source and according to the nature of the medium through which it is travelling
- 1.4 **Use the equation:**
intensity = power of incident radiation / area
$$I = \frac{P}{A}$$
- 1.5 Describe the refraction of light by converging and diverging lenses
- 1.6 Relate the power of a lens to its shape
- 1.7 Use the equation:
power of lens (dioptr, D) = 1/focal length (metre, m)
- 1.8 *Investigate variations of image characteristics with objects at different distances from a converging lens*
- 1.9 **Use the lens equation:**
$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

(f = focal length (m), u = object distance (m), v = image distance (m))
The use of the real is positive sign convention is preferred and will be used in the exam
- 1.10 Identify the following features in a diagram of the eye – cornea, iris, pupil, lens, retina, ciliary muscles
- 1.11 Demonstrate an understanding that light is focused on the retina by the action of the lens and cornea

Unit P3: Applications of physics

- 1.12 Recall that the average adult human eye has a near point at about 25 cm and a far point at infinity
- 1.13 Explain the symptoms and causes of short sight and long sight (students will not be expected to draw scaled ray diagrams, but may be expected to interpret them)
- 1.14 Compare and contrast treatments for short sight and long sight, including the use of:
 - a simple lenses
 - b contact lenses
 - c **laser correction**(combined lens equation is not required; students will not be expected to draw scaled ray diagrams, but may be expected to interpret them)
- 1.15 Explain, with the aid of ray diagrams, reflection, refraction and total internal reflection (TIR), including the law of reflection and critical angle
- 1.16 **Calculate critical angle using Snell's Law**
- 1.17 Explain refraction in terms of change of speed of radiation
- 1.18 *Investigate the critical angle for perspex/air or glass/air or water/air boundaries*
- 1.19 *Investigate TIR between different media*
- 1.20 Explain how TIR is used in optical fibres
- 1.21 Explain uses of optical fibres in endoscopes
- 1.22 Explain uses of ultrasound in diagnosis and treatment

Topic 2

X-rays and ECGs

- 2.1 Relate the ionisation by X-rays to their frequency and energy qualitatively ($E = hf$ is not required)
- 2.2 Explain the key features of passing a current through an evacuated tube, including:
 - a thermionic emission of electrons from a heated filament
 - b potential difference between the cathode (filament) and the anode (metal target)
 - c why the vacuum is necessary
 - d possible production of X-rays by collision with a metal target
- 2.3 Explain why a beam of charged particles is equivalent to an electric current
- 2.4 **Use the equation:**
current (ampere, A) = number of particles per second (1/second, 1/s) × charge on each particle (coulomb, C)
 $I = N \times q$

Unit P3: Applications of physics

2.5 Use the equation:

**kinetic energy (joule, J) = electronic charge (coulomb, C)
× accelerating potential difference (volt, V)**

$$\mathbf{KE = \frac{1}{2}mv^2 = e \times V}$$

2.6 Demonstrate an understanding of the inverse square law for electromagnetic radiation

2.7 Relate the absorption of X-rays to the thickness of the material through which they are travelling, quantitatively

2.8 Describe how X-rays are used in CAT scans and fluoroscopes

2.9 Demonstrate an understanding of the comparison of the risks and advantages of using X-rays for treatment and diagnosis

2.10 Explain how action potentials can be measured with an electrocardiogram (ECG) to monitor heart action

2.11 Relate the characteristic shape of a normal ECG to heart action

2.12 Use the equation:

frequency (hertz, Hz) = 1/time period (second, s)

$$f = \frac{1}{T}$$

2.13 Describe the use of a pacemaker to regulate the heart action

2.14 Describe the principles and use of pulse oximetry